MILITARY SPECIFICATION

# PROPELLANT GRAINS, NOSOL-318

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

**1.1 Scope.** This specification establishes requirements for two types of NOSOL-318 double-base propellants (see 6.1).

**1.2 Classification.** Propellant shall be of the following types and classes:

| Type I — Graphited | Type II — Non-graphited | |
|---|---|---|
| Class 1 | Class 1 | |
| Class 2 | Class 2 | |
| Class 3 | Class 3 | (See table I for explanation of classes) |
| Class 4 | Class 4 | |
| Class 5 | Class 5 | |

## 2. APPLICABLE DOCUMENTS

**2.1 Government documents.**

**2.1.1 Specifications and standards.** The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-G-155 | Graphite, Dry (For Use in Ammunition) |
| MIL-P-270 | Propellant, Artillery |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-286 | Propellant, Solid: Sampling, Examination and Testing |

**2.1.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (524), Indian Head, Maryland 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

DISTRIBUTION STATEMENT A. Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:24Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Naval Ordnance Systems Command (Code Ident 10001)

| | |
|---|---|
| WS 13986 | Propellant, Sheet Stock NOSOL-318 |

DRAWINGS

Naval Ordnance Systems Command (Code Ident 10001)

| | |
|---|---|
| 138439 | Packing Box Mark 7 (For Smokeless Powder), General Arrangement |
| 138440 | Packing Box Mark 7 (For Smokeless Powder), Body Details |
| 138441 | Packing Box Mark 7 (For Smokeless Powder), Cover Details |

Naval Ordnance Station, Indian Head, Maryland (Code Ident 14083)

| | |
|---|---|
| 5755 | Leverpak Fibre Drum |

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

| | |
|---|---|
| OD 40816 | Propellant Density, Mercury-Macro Method |

Code of Federal Regulations

| | |
|---|---|
| 49 CFR 100 - 199 | Transportation |

Naval Ordnance Station, Indian Head, Maryland (Code Ident 14083)

| | |
|---|---|
| IHSP 85-208 | YBOMB: A Computer Program for the Collection, Storage, and Processing of Pressure-Time Data from Propellant Burned in a Closed Bomb |

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

**2.2 Order of Precedence.** In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

**3.2 Materials.**

**3.2.1 Sheet stock.** Sheet stock propellant conforming to WS 13986 shall be used in the manufacture of the propellant grain. No more than five lots of sheet stock shall be used to manufacture one lot of propellant.



Source: https://assist.dla.mil -- Downloaded: 202610-23T07:24Z
Check the source to verify that this is the currentversion before use.

**3.2.2 Graphite.** Graphite conforming to MIL-G-155, Grade IV, shall be used to glaze the Type I granulations.

**3.3 Product characteristics.**

**3.3.1 Grain form.** Granulations shall be either graphited (Type I) or non-graphited (Type II), as specified in the contract. Each type shall consist of five classes as specified in table I and shown in figure 1.

TABLE I. Granulation characteristics by class.

| Class | Length (in.) | Diameter (in.) | Inner Web (in.) | Outer Web (in.) | Perf Diameter (in.) |
|---|---|---|---|---|---|
| 1 (7 perf, tall) | .670 nominal | .325 nominal | .054 nominal | .054 nominal | .036 nominal |
| 2 (flake) | .010 nominal | .050 nominal | Not applicable | Not applicable | Not applicable |
| 3 (giant) | .30 ± .01 | .43 ± .01 | Not applicable | Not applicable | Not applicable |
| 4 (midget) | .10 ± .01 | .08 ± .03 | Not applicable | Not applicable | Not applicable |
| 5 (7 perf, short) | .375 nominal | .630 nominal | .0825 nominal | .0825 nominal | .100 nominal |

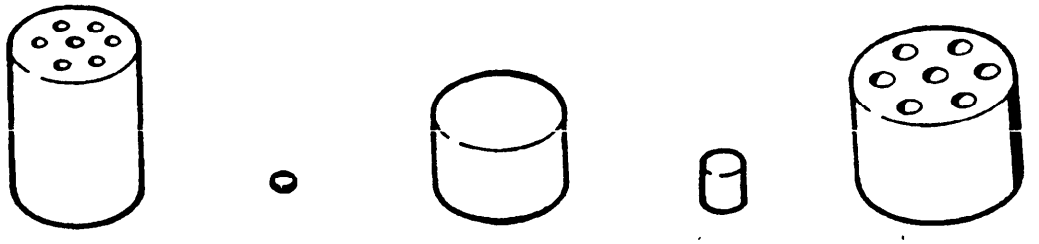



Figure 1. Propellant grain classes.

**3.3.2 Grain dimensions.** Grain dimensions shall be as specified in table I (see 4.6.1).

**3.3.3 Grain integrity (Class 1 and 5 only).** The propellant grains shall be homogeneous, without cracks in inner web or outer web (see 4.6.2).

**3.3.4 Graphite content.** The graphite content of the propellant grains shall be as specified in the contract (see 4.6.3 and 6.2.1).

**3.3.5 Moisture content.** The moisture content of the propellant grains shall be not greater than 0.5 percent by weight (see 4.6.4).

**3.3.6 Specific gravity.** Specific gravity shall be not less than 1.48 grams per cubic centimeter (see 4.6.5).

**3.3.7 Stability.** The slope at 100 minutes shall be not greater than 1.00 millimeters per minute when the propellant is subjected to Taliani testing (see 4.6.6). If the 120° C heat test is used (see 4.6.6.1), the methyl violet test paper shall not fade to a salmon-pink color in less than 2 hours.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:24Z
Check the source to verify that this is the current version before use.

**3.3.8 Delivered impetus.** Delivered impetus shall be not less than 235,200 ft. lb./lb. nor greater than 277,900 ft. lb./lb. (see 4.6.7).

**3.4 Workmanship.** The propellant grains shall be uniform in quality and free from foreign material.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.1.1 Responsibility for compliance.** All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

**4.1.2 Test report.** When specified in the contract (see 6.2.1), the contractor shall furnish test reports to the procuring activity.

**4.2 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

**4.3 Inspection conditions.** Unless otherwise specified (see 6.2.1), all inspections shall be performed as specified in 4.6.

**4.4 First article inspection.** First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2.1). First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production.

**4.4.1 Sample size.** The first article sample shall consist of 10 pounds of propellant grains.

**4.4.2 Inspection routine.** The sample shall undergo the tests specified in 4.6 and any additional tests as specified by the procuring activity.

**4.4.3 Failures.** Failure of the propellant grains to meet any of the above requirements shall be cause for first article rejection.

**4.5 Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in 4.6. Failure of any sample to conform to any requirement specified herein shall be cause for rejection of the lot represented.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:24Z
Check the source to verify that this is the current version before use.

**4.5.1 Lot size.** Unless otherwise specified in the contract or purchase order, the propellant shall be produced in lots of 100,000 pounds.

**4.5.2 Sampling.** The samples shall be selected from containers so as to be representative of the lot.

**4.5.2.1 For chemical and stability tests.** A 10-pound sample shall be selected from each lot. An additional 10-pound sample shall be selected and forwarded to the Naval Ordnance Station, Indian Head, Md. 20640 (see 6.2.1).

**4.6 Test Methods.**

**4.6.1 Grain dimensions.** Grain dimensions shall be determined in accordance with MIL-STD-286, method 504.1.1

**4.6.2 Grain integrity.** The absence of cracks or incipient cracks shall be verified by the use of a magnifying instrument, such as a microscope or optical comparator, giving a magnification of at least 20X. The lighting used for the test shall be transmitted, not incident, through the sample. The samples shall be sections taken from the approximate midpoint of the grains, with a thickness of 10 microns or less.

**4.6.3 Graphite determination.** The percent by weight of graphite shall be determined in accordance with MIL-STD-286, method 309.1.2.

**4.6.4 Moisture determination.** The moisture content shall be determined in accordance with MIL-STD-286, method 103.1.3.

**4.6.5 Specific gravity determination.** Specific gravity shall be determined in accordance with OD 40816.

**4.6.6 Stability.** Stability shall be determined in accordance with MIL-STD-286, method 406.1.2.

**4.6.6.1 Alternate method.** Method 404.1.2 (120° C heat test) of MIL-STD-286 may be used to determine stability.

**4.6.7 Delivered impetus test.** The delivered impetus shall be determined as specified in MIL-STD-286, Method 801.1 for closed bomb method and IHSP 85-208 for computer determination.

**4.6.8 Visual examination.** The propellant grains shall be visually examined to verify the following:

a. Type I grains are completely coated with graphite.
b. The grain surfaces are smooth.
c. The grains conform to the requirements of 3.4.

## 5. PACKAGING

**5.1 Packing.** Unless otherwise specified (see 6.2.1), packing shall be level C for Class 2 and Class 4 granulations. Class 2 and 4 shall be limited to 25 pounds (maximum) per pack. Packing for other classes shall be level A or C, as specified in the contract or purchase order.

**5.1.1 Level A.** The propellant shall be packed in containers conforming to Drawings 138439, 138440, and 138441 and in accordance with the packing requirements of MIL-P-270 and 49 CFR 170-190. MIL-P-270 shall apply only for Class 1 and Class 5 granulations.



Source: https://assist.dla.mil -- Downloaded: 202610-23T07:24Z
Check the source to verify that this is the currentversion before use.

**5.1.2 Level C.** The propellant grains shall be packed in fibre drums conforming to NOS Drawing 5755. The drum lid and ring closure device shall be completely interchangeable with all drums supplied. The exterior of the drum, except top and bottom, shall be painted with pearl gray alkyd base paint. Two warning signs shall be applied on opposite sides of each drum approximately 6 inches from the top by means of either pressure sensitive labels or by silk screen printing. Sign shall be 5 by 8 inches with red background and pearl gray lettering and read "WARNING-DO NOT EXPOSE TO HEAT OR DIRECT SUNLIGHT". Lettering for the word "WARNING" shall be 1 inch high; all other lettering shall be one-half inch high.

**5.2 Marking.** In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of MIL-STD-129, 49 CFR 170-190, and the following:

| | |
|---|---|
| Class 2 | DOT Propellant, Explosive, Class A |
| Class 1, 3, 4, and 5 | DOT Propellant, Explosive, Solid, Class B |

## 6. NOTES

**6.1 Intended use.** The propellant covered by this specification is intended for use as a propelling charge in ordnance devices.

**6.2 Ordering data.**

**6.2.1 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type, class, and quantity of propellant (see 1.2)
c. First article sample, if applicable, and test facility where sample is to be shipped for evaluation (see 3.1 and 3.4)
d. Graphite content of propellant grains (see 3.3.4)
e. Test reports, if applicable (see 4.1.1)
f. Inspection conditions, if other than as specified (see 4.3)
g. Any additional tests (see 4.4.2)
h. That a sample is to be furnished to Naval Ordnance Station, Indian Head (see 4.5.2.1)
i. Level of packing (see 5.1)
j. Additional markings, if required (see 5.2)
k. That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable.

**6.2.2 Data requirements.** When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

| Paragraph no. | Data requirement title | Applicable DID no. |
|---|---|---|
| 4.1.2 | Reports, Test | DI-T-2072 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:24Z
Check the source to verify that this is the current version before use.

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

**6.3 First article.** When a first article inspection is required, the item should be a first article sample. The first article should consist of 10 pounds of propellant grains. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Preparing activity:
Navy -OS
Project 1377-N764



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:24Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:24Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Ordnance Station
Standardization/Documentation Div. (524)
Indian Head, MD 20640 -5000

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Ordnance Station
Standardization/Documentation Div. (524)
Indian Head, MD 20640-5000

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:24Z
Check the source to verify that this is the currentversion before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-P-82739(OS) | PROPELLANT GRAINS, NOSOL-318 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:24Z
Check the source to verify that this is the current version before use.